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Weekly update 1 of 5 · week of Nov 9

Data, ground truth and experiment setup

THIS UPDATE IN THREE NUMBERS

292

buildings with an enclosure A1–A3 reference value (Benke et al. 2025)

78

median $kgCO_2e/m^2$ floor area; range 0.5–514

15.5%

of all enclosure carbon comes from XPS insulation alone

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Ground truth: enclosure A1–A3 per m^2 of floor area

55,807

material rows: Shell – Enclosure, stage A1–A3

243 + 49

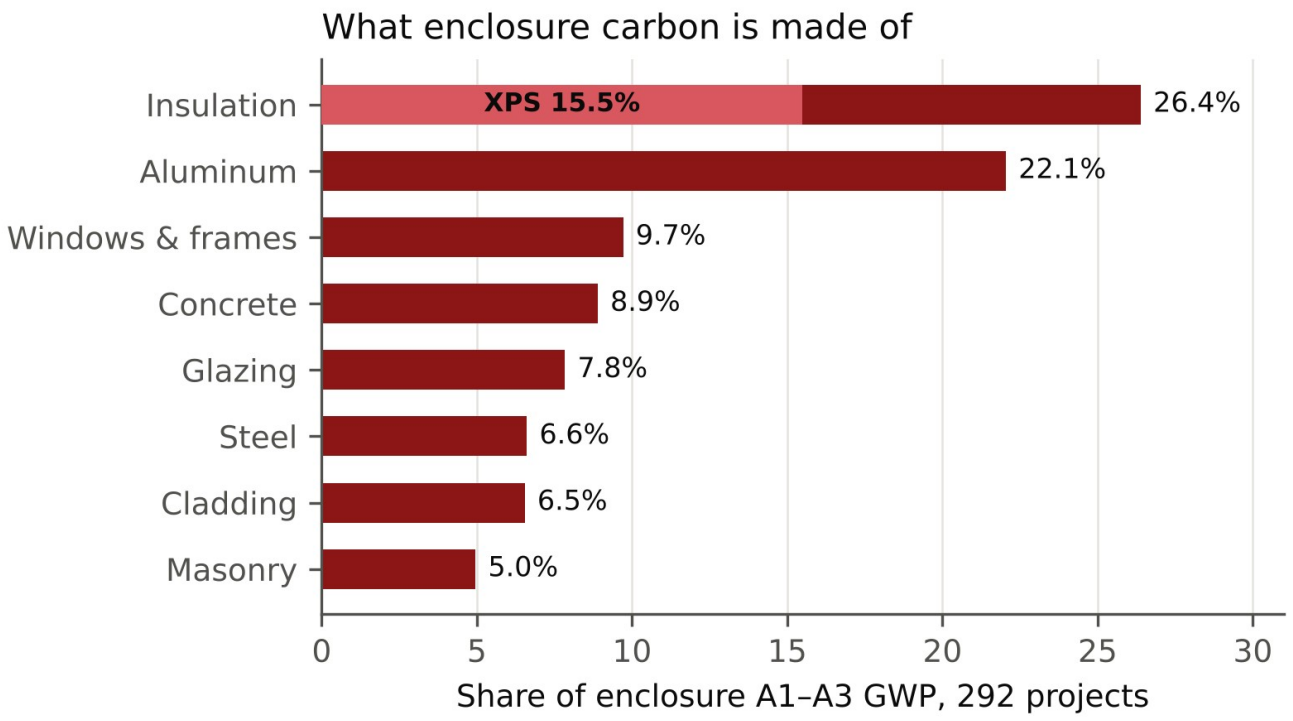
new construction + renovations

19%

median share of whole-building A–C carbon

- **Source:** harmonized North American WBLCA dataset (Benke et al. 2025, CC BY 4.0), 48 early-stage features per project.
- **Normalization:** conditioned floor area; renovated + added GFA for renovations. This reproduces the dataset's own normalization for all 292 projects.
- **Scope:** the enclosure includes windows, doors and roofs. The unit is per m² of floor area, because wall area is missing for most projects (the proposal planned per m² of wall).

What enclosure carbon is made of



- **Insulation (26.4%) and aluminum (22.0%)** dominate.
- **XPS: 43.6 kgCO_2e/kg**, against 1.2 for mineral wool and 3.1 for polyiso: the HFC-blown product that Tally and One Click LCA modelled in the assessment years.
- **149 of 292** projects contain XPS.
- **Data quality:** 10 new-construction projects have no fenestration GWP at all, several with a WWR of 0.5–0.7. They are flagged, and results are reported with and without them.

Climate vs use: use matters more (proposal RQ1)

53.6

median $kgCO_2e/m^2$, multifamily (44 projects)

87.2

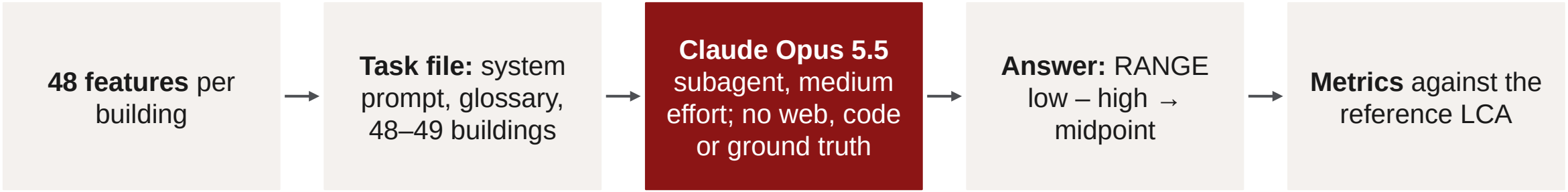
median, office (41)

157.2

median, transportation hubs (9)

- **Use differs significantly** across 9 uses (Kruskal–Wallis $p = 2.7 \times 10^{-7}$). New construction with complete references, 233 projects.
- **Climate does not reach significance** ($p = 0.26$): warm zones 80.6, zone 4 85.9, cold zones 95.1.
- Per m^2 of floor area, use (glazing share, envelope-to-floor ratio) shapes enclosure carbon more than climate (insulation).

LLM experiment protocol



- **3 runs × 292 buildings** = 876 estimates per prompt. Seeded shuffles; identical chunks across prompt versions, so comparisons are paired building by building.
- **Metrics:** ±20% coverage (Wilson CI), RMSE, median APE, median |log(est/ref)|, bias (geometric mean of est/ref), CV across runs.
- **Parser change:** a lower bound of 0 is accepted. Renovations that leave the envelope untouched have reference values near zero, and the course parser would have dropped exactly those answers.

Next update

Coming next

- **Baseline results** with a plain prompt (the model's own knowledge, prompt v1).
- **Error analysis:** where the model under- and overestimates, and why.

Input we'd like

- Is **per m² of floor area** acceptable as the functional unit, instead of per m² of wall?
- Keep the **10 flagged incomplete references** in the main metrics (we report both ways)?

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Weekly update 2 of 5 · week of Nov 16

Baseline (prompt v1) and error analysis

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19.5%

of estimates fall within ±20% of the reference LCA

0.75

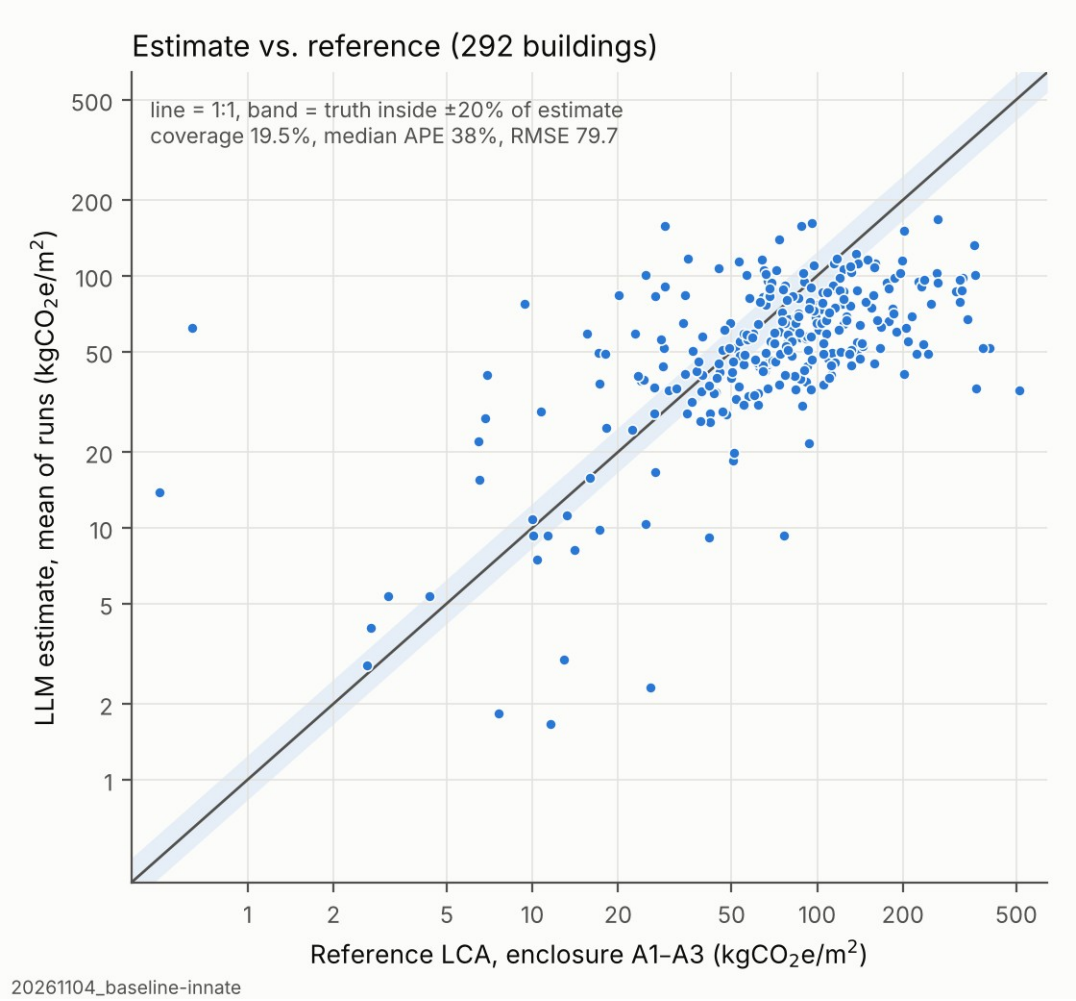
bias: estimates are about 25% low on average

0.46

log–log slope: every building is pulled toward a typical value

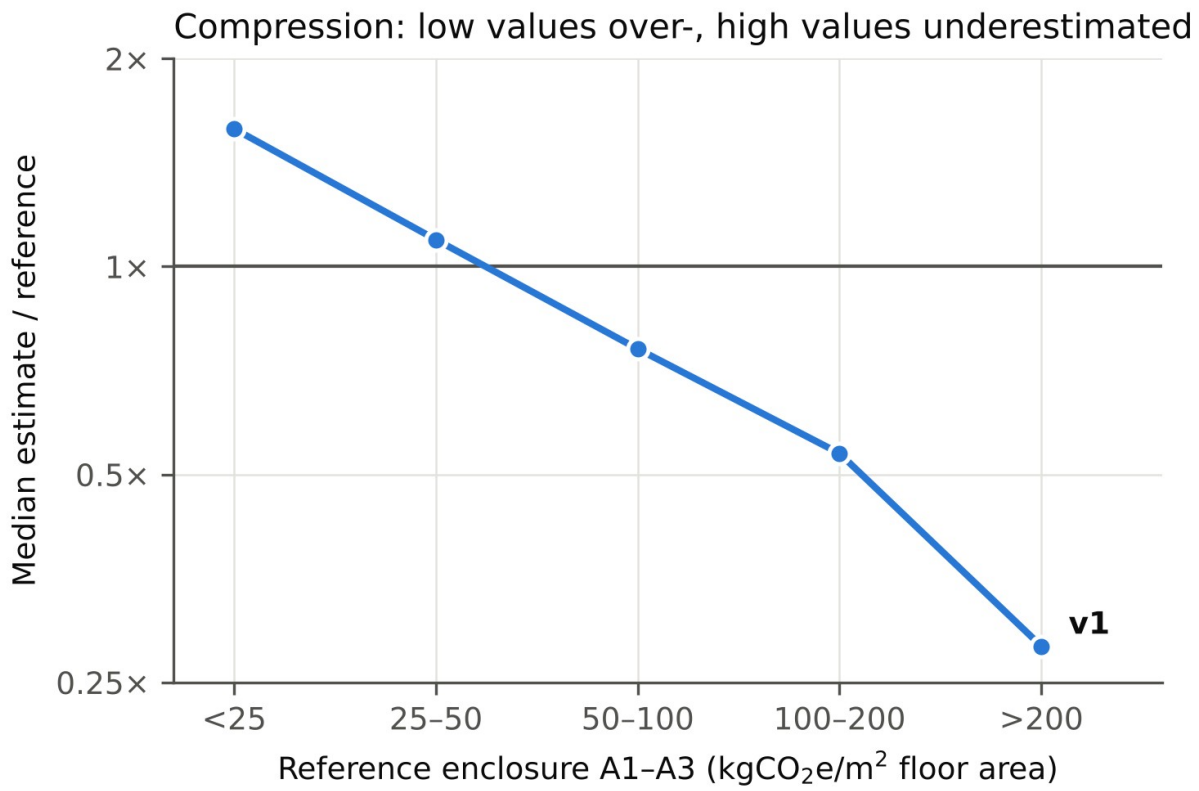
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Baseline: a plain prompt, the model's own knowledge



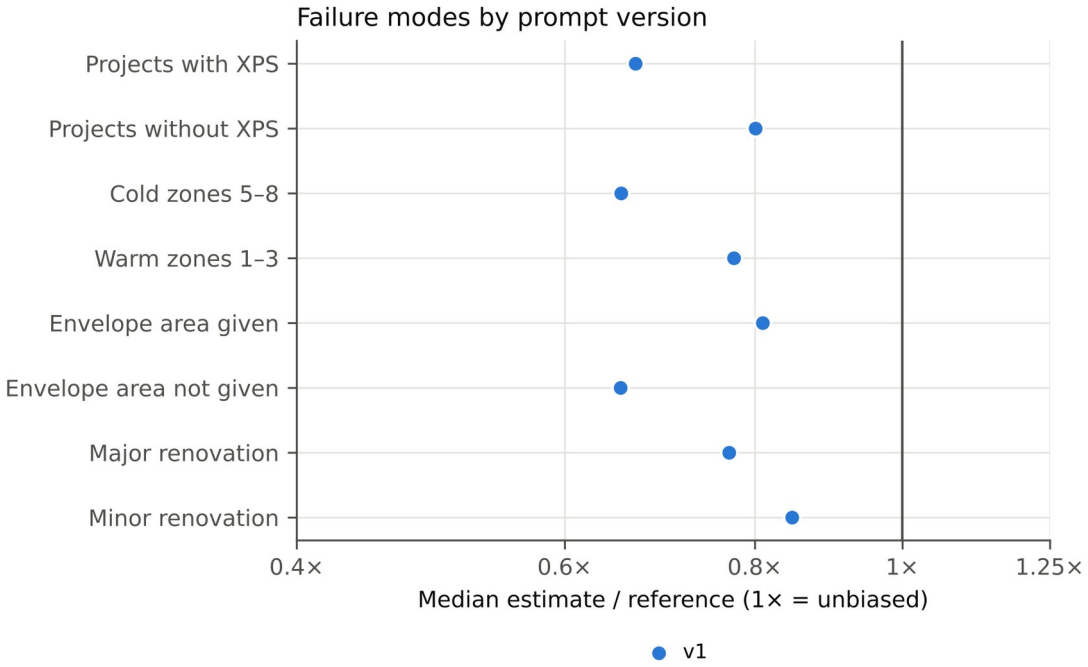
- **Coverage 19.5%** (CI 15.4–24.5%), RMSE 79.7, median APE 37.9%, CV 19.0% across runs.
- **63%** of buildings are underestimated beyond the band; 17.5% are overestimated.
- The **20 largest errors**, all underestimates, carry 68% of the squared error.

Main pattern: compression, not uniform underestimation



- Below 25 $kgCO_2e/m^2$: overestimated **1.58×**. Above 200: estimated at **0.28×**.
- The model leans on **geometry**: the envelope-to-floor ratio correlates 0.62 with the log estimate but only 0.39 with the log reference.
- What it misses is **carbon per m² of envelope** ($r = -0.79$ with the log error). That is set by assembly and material choices, and **no feature describes the wall assembly**.

Why the model underestimates



1. **HFC-blown XPS is never considered:** 0.6% of answers mention XPS. Ratio 0.67 with XPS vs 0.80 without; removing XPS from the reference moves 0.72 → 0.84, about **half the bias**.
2. **The climate gap is an insulation effect:** cold zones 0.65, warm 0.77. XPS is in 65–68% of zone 5A/5B projects but 26–33% of zone 3B/3C.
3. **Missing envelope area:** 0.81 when given, 0.65 when the model must estimate it.
4. **Curtain-wall tail and renovations:** single projects reach 300 from curtain wall alone; 60% of major renovations are underestimated.
5. **Scope wording** was narrower than the dataset's enclosure (11% of enclosure GWP is tagged "Superstructure" in the source tool).

Why the model overestimates

3.09×

median estimate/reference for the 10 incomplete references

10

projects with no fenestration GWP, flagged

Fit-outs

and minor renovations: envelope scope unknown from the features

- The **largest overestimates trace back to incomplete reference LCAs**, not to the model.
- The remaining overestimates are mostly fit-outs and minor renovations, where the extent of envelope work is not described by any feature.
- Flagged projects: 27, 35, 66, 74, 80, 91, 141, 217, 234, 264.

Next update

Coming next

- **Prompt v2:** add the missing knowledge. A scope matching the dataset, a component take-off, named high-carbon materials, and a renovation rule.
- Rerun 3 × 292 and compare **paired** with v1.

Input we'd like

- Feedback on the **error taxonomy** before we revise the prompt.
- Any **domain knowledge** you would add to the prompt?

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Weekly update 3 of 5 · week of Nov 23

Adding knowledge: v2 and documents

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13.8%

run-to-run CV (v1 19.0%): more consistent

15.1%

coverage (v1 19.5%): less accurate

0.61

bias with envelope documents in context (v1 0.72): worse

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What prompt v2 changed

1 Scope

Written to match the dataset's enclosure definition.

2 Component take-off

Areas → assemblies → intensities → total.

3 High-carbon materials

HFC-blown XPS "ten times or more" the GWP of other insulation per unit R; aluminum curtain wall 100–200 $kgCO_2e/m^2$.

4 No anchoring

Estimating from a "typical building" is ruled out.

5 Renovation rule

Count only the materials of the envelope work.

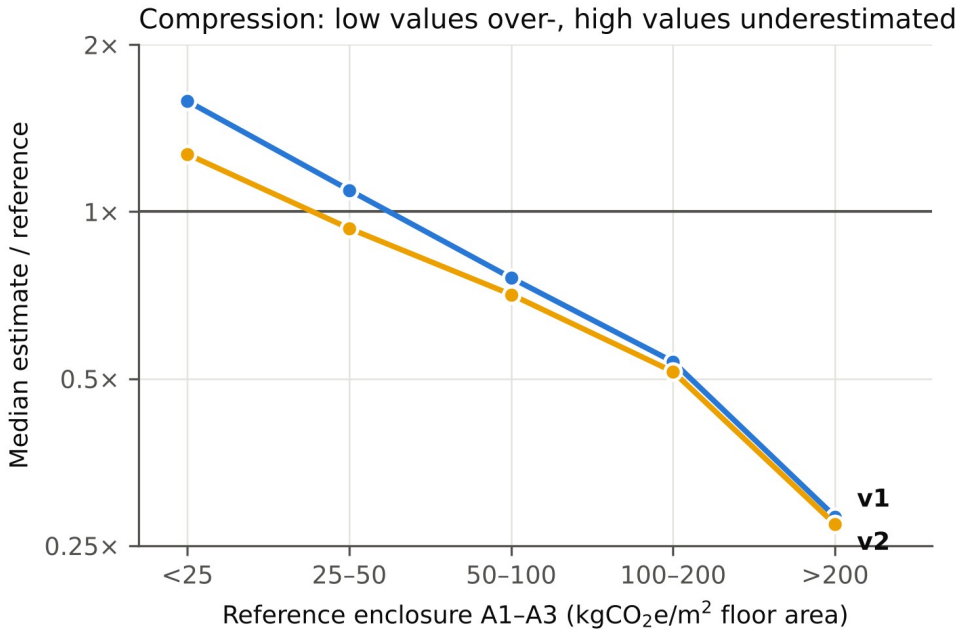
6 BEST value

A best estimate is requested alongside the range.

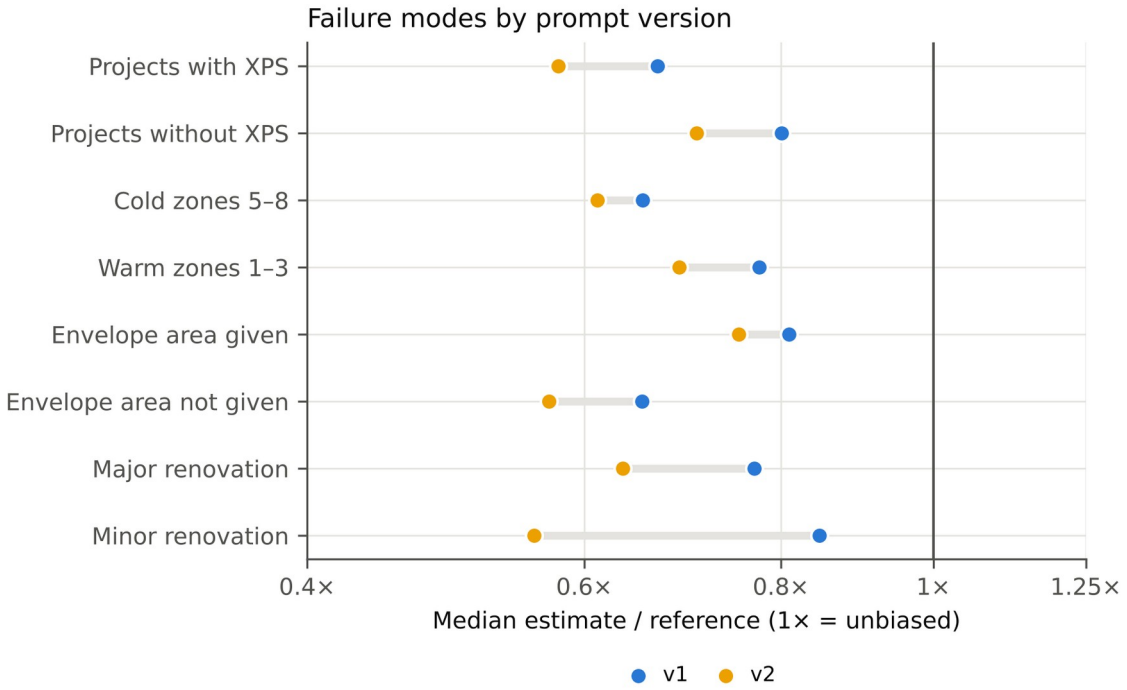
Results: v2 vs v1 (292 buildings, 3 runs each)

Prompt	Coverage	RMSE	Med. APE	Med. |log|	Bias	CV
v1	19.5%	79.7	37.9%	0.456	0.75	19.0%
v2	15.1%	81.0	44.3%	0.563	0.67	13.8%

- v2 errors are **significantly larger** (Wilcoxon $p < 0.0001$).
- Compression eased (slope 0.46 → 0.55), but the **whole curve shifted down**.
- More consistent across runs: CV 19.0% → 13.8%.



Why v2 got worse



1. **The facts were repeated, not applied.** XPS was named 50× more often, but 89% of answers still assumed polyiso roofs; the median roof intensity stayed at 35 $kgCO_2e/m^2$.
2. **The take-offs were too light.** Bottom-up intensities came out at 66.8 per m^2 of envelope, against 86.7 in the reference.
3. **A subtracting rule hurt.** "Count only new materials" dropped minor renovations from 0.85 to 0.56.

Second way to add knowledge: envelope documents in context

Condition (run 0)	Coverage	Med. |log|	Bias	RMSE
v1 (innate)	17.8%	0.519	0.72	81.2
v1 + 4 envelope docs	15.8%	0.609	0.61	85.3

- Docs: RDH 2025 wall assemblies, CLF 2023 material baselines, Arup facades/glass, UBC fenestration (30k characters each); v1 task text unchanged.
- **Worse:** log error +0.090 (Wilcoxon $p < 0.001$); bias 0.72 → 0.61.

Why

- **The documents were read:** 22% of answers cite them, mostly for the low end of a range.
- But they describe **today's low-carbon products** (HFO foams, recent EPDs, modern glazing); the reference LCAs used period tool data.
- XPS mentions fell to **0%**; the XPS-project ratio fell 0.64 → 0.50.

Lesson for prompt design

Qualitative knowledge, as instructions or as documents, gets repeated, not applied. To move the numbers, give numbers, at the step where the model chooses them.

This is the design principle behind **prompt v3**.

Next update

Coming next

- **Prompt v3:** sourced quantitative anchors. A table of external A1–A3 intensities per m² of component, placed at the decision step.
- **Held-out check** on buildings not used for the prompt diagnostics.

Input we'd like

- Sources for intensities: CLF baselines, PCI and Kawneer EPDs, RDH studies. **None from the ground-truth dataset or the CLF WBLCA benchmark** (leakage). Any to add or avoid?

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Weekly update 4 of 5 · week of Nov 30

Prompt v3 and a held-out check

THIS UPDATE IN THREE NUMBERS

25.0%

coverage (v1 19.5%, v2 15.1%)

0.86

bias (v1 0.75): half of the underestimate removed

12.5%

run-to-run CV (v1 19.0%)

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What prompt v3 adds (on top of the v2 scope and method)

1 Sourced intensity table

A1–A3 per m² of component, compiled by a separate research subagent from external sources (CLF 2025 baselines, PCI, Kawneer, RDH); **excludes the ground-truth dataset**. Placed at the step where numbers are chosen.

2 XPS modelling convention

LCA tools of the period often used **HFC-blown XPS** for roofs.

3 Sanity check

Roughly **50 kgCO_2e/m² of envelope** as a cross-check of the total.

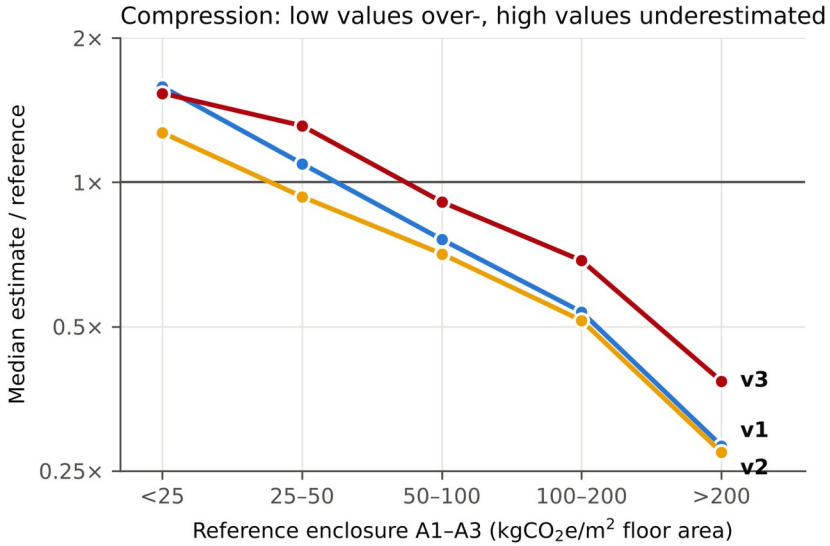
4 Renovation rule

Rewritten so that it **no longer subtracts** by default.

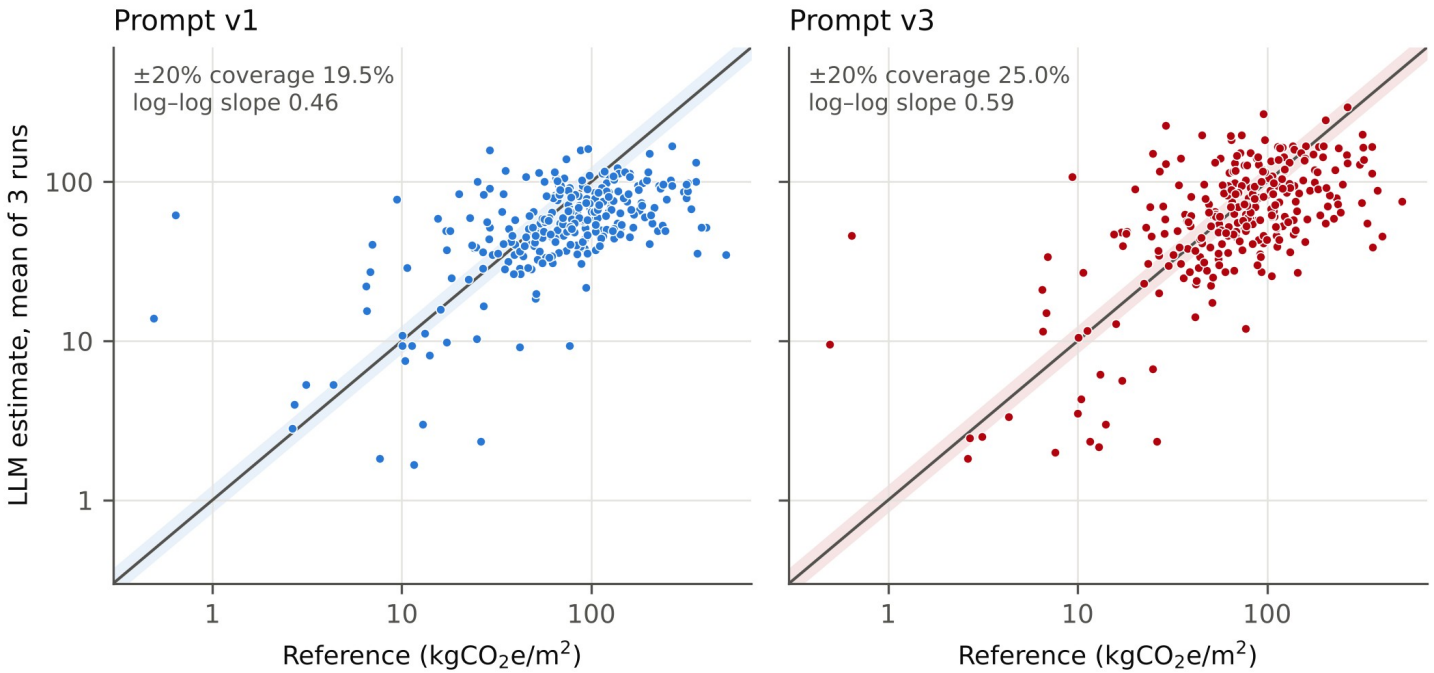
Results: all three prompts (292 buildings, 3 runs each)

Prompt	Coverage	RMSE	Med. APE	Med. |log|	Bias	CV
v1	19.5%	79.7	37.9%	0.456	0.75	19.0%
v2	15.1%	81.0	44.3%	0.563	0.67	13.8%
v3	**25.0%**	**76.0**	**40.0%**	**0.426**	**0.86**	**12.5%**

- v3 vs v2: **significant** (p = 0.002). v3 vs v1: consistent but **borderline** (McNemar p = 0.085, Wilcoxon p = 0.053).
- HFC-XPS is now considered in **96%** of answers for projects with XPS.
- Unbiased now: the non-XPS part (0.99), buildings with a given envelope area (0.99), warm zones (0.98).

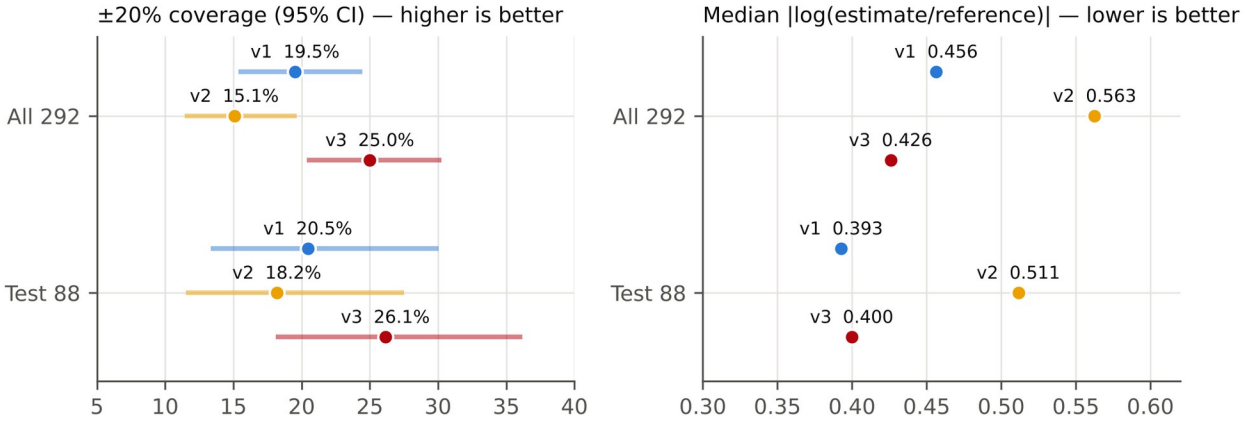


What is still wrong



- **Compression at the top:** 0.38 above 200 $kgCO_2e/m^2$.
- **Light enclosures** are overestimated more (1.31 in the 25–50 bin).
- **Renovations** remain hard: the extent of envelope work is not in the features.

Held-out check: 88 test buildings



- **No external test set fits:** the CLF benchmark probably shares source models; CarbEnMats has 172 European buildings, a different element definition and no geometry.
- **Split:** 88 test / 204 dev, stratified on project type × climate (seed 226). It was **assigned after the runs**, so we re-ran every diagnostic on dev only, and all findings replicate (slope 0.44, XPS 14.7%, area effect 0.79 vs 0.65, polyiso 89%).
- **Test:** v3 26.1% vs v1 20.5% coverage, RMSE 69.2 vs 74.1, bias 0.86 vs 0.75. Median |log| unchanged (0.400 vs 0.393); **not significant** at this size (p = 0.41).

Next update

Coming next

- **Feature study:** which inputs does the model actually need?
- **LEED v5:** what the accuracy means for MRp2 and MRc2.
- Final deliverables: paper, final prompt, poster.

Input we'd like

- Is the **post-hoc split with dev-only replication** acceptable, or should we collect additional projects?

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Weekly update 5 of 5 · week of Dec 7

Features, LEED v5 and deliverables

THIS UPDATE IN THREE NUMBERS

10

early-design features are as good as all 48

7×

prompt content moved the error 7× more than any feature set

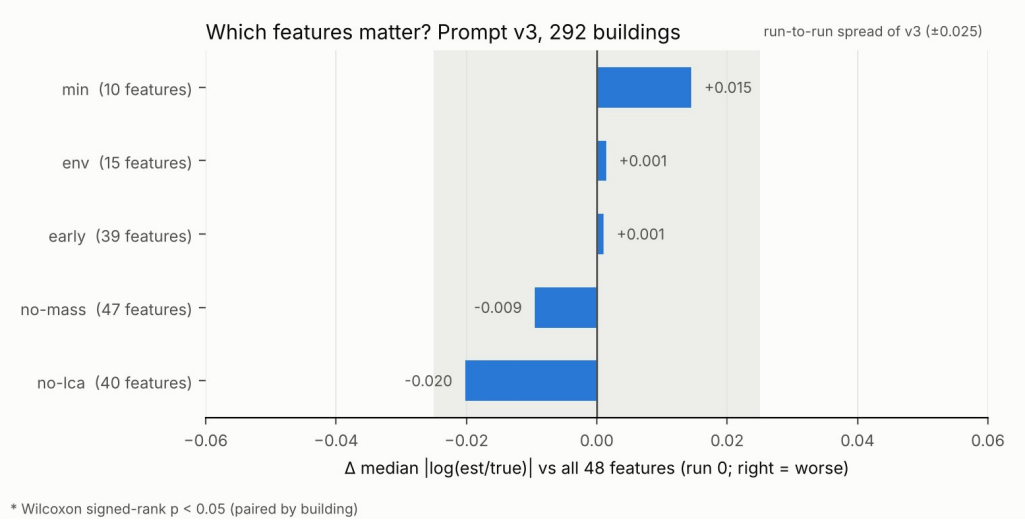
≈ 6%

of whole-building carbon from a 30% enclosure cut

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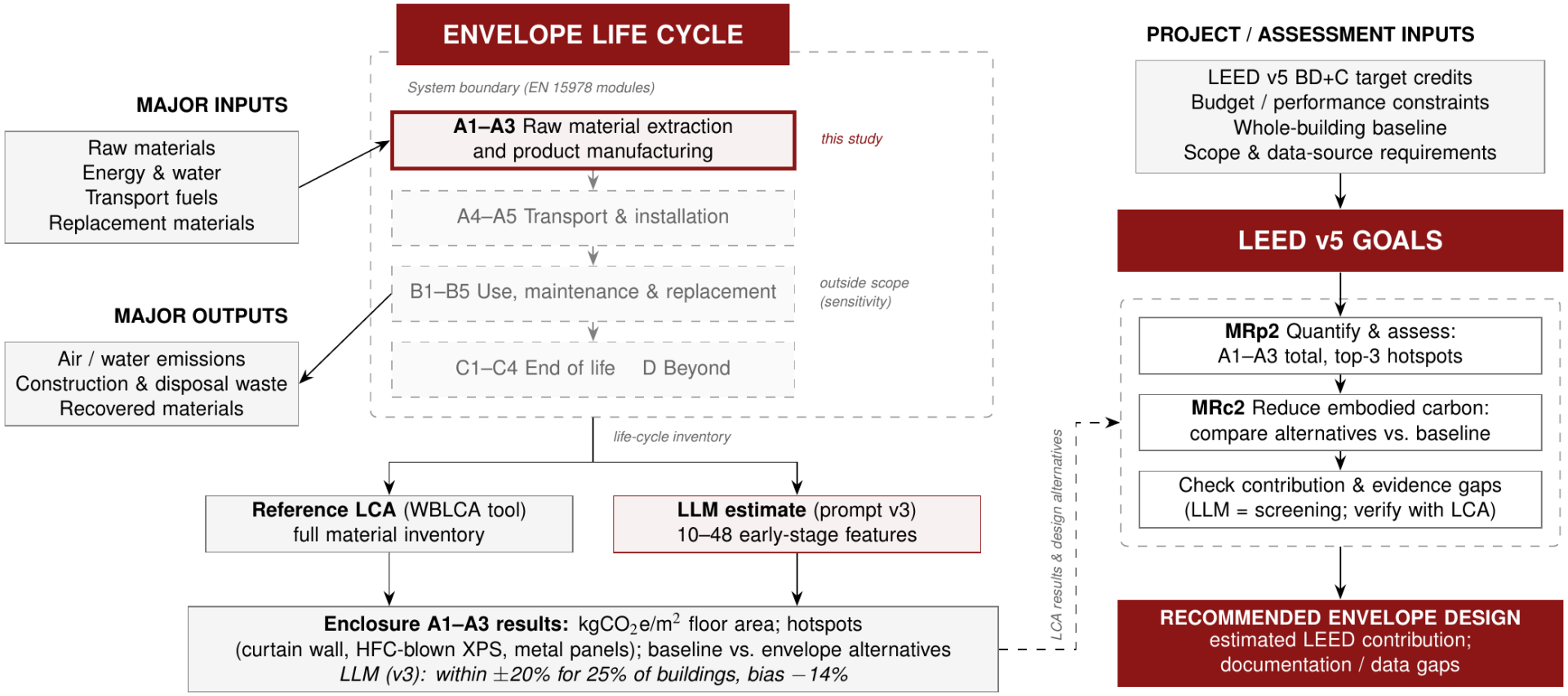
Feature study: 10 inputs are enough (v3, run 0)

Set	n	Coverage	Med. |log|	Δ	p	Bias
all	48	24.3%	0.461	–	–	0.86
early	39	21.6%	0.462	+0.001	0.13	0.88
env	15	19.9%	0.463	+0.001	0.33	0.93
min	10	25.3%	0.476	+0.015	0.81	0.81
no-lca	40	25.3%	0.441	−0.020	0.57	0.90
no-mass	47	22.6%	0.452	−0.009	0.92	0.85



- No set differs significantly from all 48; every difference is within v3's run-to-run spread (±0.025).
- The model does **not depend on mass_total** or other LCA-derived fields.
- One run per set; 7 of 9 planned ablations were stopped to limit cost.

Where the estimate fits in LEED v5



The study boundary is A1–A3. A reference LCA and the LLM estimate both produce enclosure results that feed **MRp2** (quantify, hotspots) and **MRc2** (reduce against a baseline). The LLM screens; an LCA documents.

LEED v5: screening yes, credit documentation no

1 MRp2: hotspots

Answers name the reference **top-1 hotspot in 95%** of cases (chance at the same count: 61%), but list ~7 categories, not a ranked top 3. Bias is small (−14%). **Suited to screening; ask for a ranked top 3.**

2 MRc2: reductions

CV 12.5%, median error ≈ 35–40%, coverage 1 in 4: as large as typical documented reductions. **Cannot document a credit-level reduction.** Paired comparison of alternatives is the promising, untested use.

3 Option 1: dilution

The enclosure is a median **19%** of whole-building A–C, so a 30% enclosure cut ≈ **6%** of the building. Envelope choices work best together with structure, or via Option 2 EPDs.

Deliverables and limitations

1 Final prompt (v3)

Full text in the paper's appendix, with the sourced intensity table.

2 Paper, reports, poster

ICML-style paper, English and Chinese reports, PPT poster; all code and results in the GitHub repo.

3 Limitations

Answers in one task file share a context; post-hoc split; v3–v1 gain borderline with 3 runs; feature study and document condition one run each; per m² floor area; uneven reference quality; one model.

4 Future work

Assembly-type inputs; paired alternative comparison; a ranked top-3 hotspot output for MRp2; an external test set.

Wrap-up

Take-aways

- An LLM's error is **structured**: compression and material blind spots, which error analysis can locate.
- **Sourced numbers at the decision step** work; qualitative knowledge does not.
- For LEED v5: **screening and comparison**, verified by an LCA.

Input we'd like

- Feedback on the paper and poster before final submission.
- Interest in the **assembly-type** follow-up?